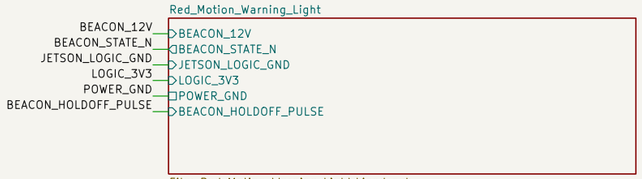
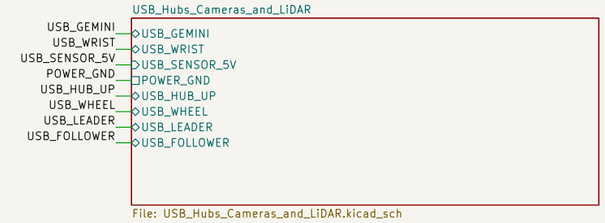
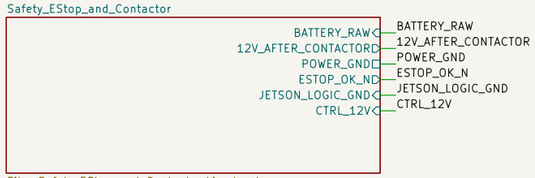
KIRO Wiring V2 | System_Overview

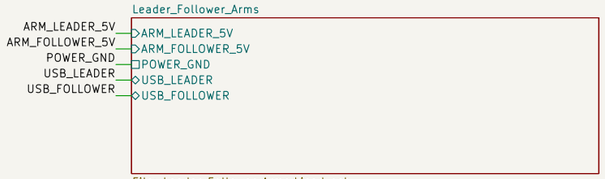
Battery-powered KIRO robot: 12 V 45 Ah battery, Jetson Orin Nano Super, 4-wheel mecanum (ST3215), LeRobot leader + follower arms, 2 cameras, 1 LiDAR, mushroom E-stop + contactor, red motion-warning light

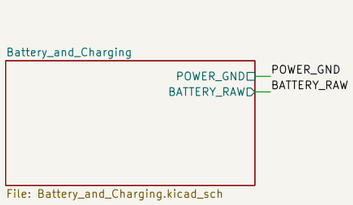


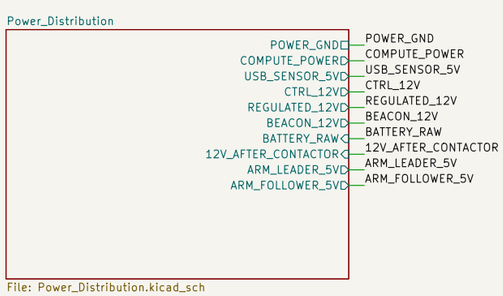


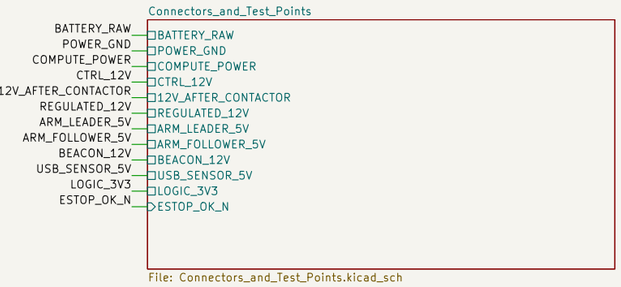














POWER DOMAINS

BATTERY_RAW ..... battery after F_MAIN and SW_MAIN (10.0–14.6 V worst case) not cut by E-stop

COMPUTE_POWER ..... F_JETSON -> ideal-diode -> Jetson J16 not cut by E-stop

USB_SENSOR_5V ..... F_HUB -> hub PSU -> powered USB hub (cameras/LiDAR on USB) not cut by E-stop

CTRL_12V ..... F_CTRL -> mushroom NC -> START / K1 AUX -> K1 coil E-stop loop

12V_AFTER_CONTACTOR . K1 main contact output = DRIVE_ARMED cut by E-stop

DRIVE_POWER / REGULATED_12V .. F_DRIVE -> 12 V stabiliser -> wheel adapter cut by E-stop

ARM_POWER ..... F_ARM_IN -> EA75-5V -> F_LEADER / F_FOLLOWER cut by E-stop

BEACON_12V ..... F_BEACON -> red light, low-side MOSFET cut by E-stop

LOGIC_3V3 / JETSON_LOGIC_GND .. Jetson J12 pin 1 / pin 6 (buffers only) LOGIC_5V (J12 pin 2): unused

POWER_GND ..... single star point at battery negative

Signals to/from Jetson: ESTOP_OK_N (in, LOW = armed), BEACON_STATE_N (in, LOW = light on), BEACON_HOLDOFF_PULSE (out, >=100 Hz = light off; stopped = MOTION_WARNING_REQUEST).

V2 2026-11-11. Values marked (GEN) are generic placeholders until real part data is supplied — see engineering document section 15.

Sheet: /
File: kiro_wiring_v2.kicad_sch

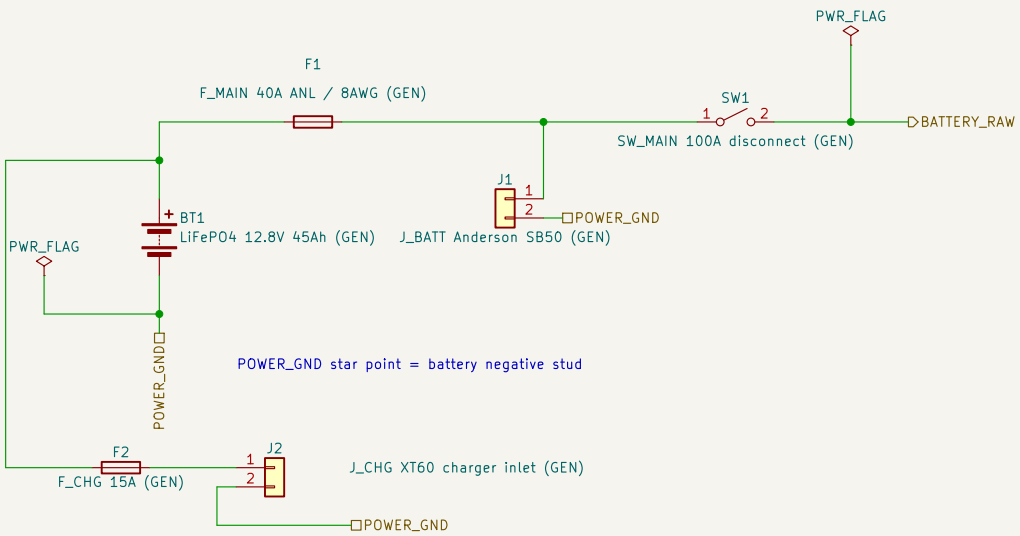
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KIRO Wiring V2 | Battery_and_Charging

Battery, main fuse, disconnect, charger inlet and POWER_GND star point



BATTERY — GENERIC ASSUMPTION (replace with label data)

Chemistry ..... LiFePO4 4S (GEN)

Nominal voltage ..... 12.8 V

Full-charge voltage .. 14.6 V

Minimum safe voltage . 10.0 V (BMS cut-off)

BMS ..... 50 A continuous / 100 A peak (GEN)

Charger ..... LiFePO4 14.6 V 10 A (GEN)

Main connector . Anderson SB50, red = + (GEN)

Design worst case raw range: 10.0 – 14.6 V

Energy 576 Wh nominal; ~461 Wh usable (80 %, estimate)

J2/F2: charger joins on the battery side of SW_MAIN, so the robot can charge while switched off.

Reverse polarity: J_BATT must be keyed; compute branch adds an ideal-diode stage (Power_Distribution).

F_MAIN within 150 mm of BT1+. 40 A: above 25.5 A peak load, below 50 A BMS and 8 AWG rating.

Sheet: /Battery_and_Charging/
File: Battery_and_Charging.kicad_sch

Title:

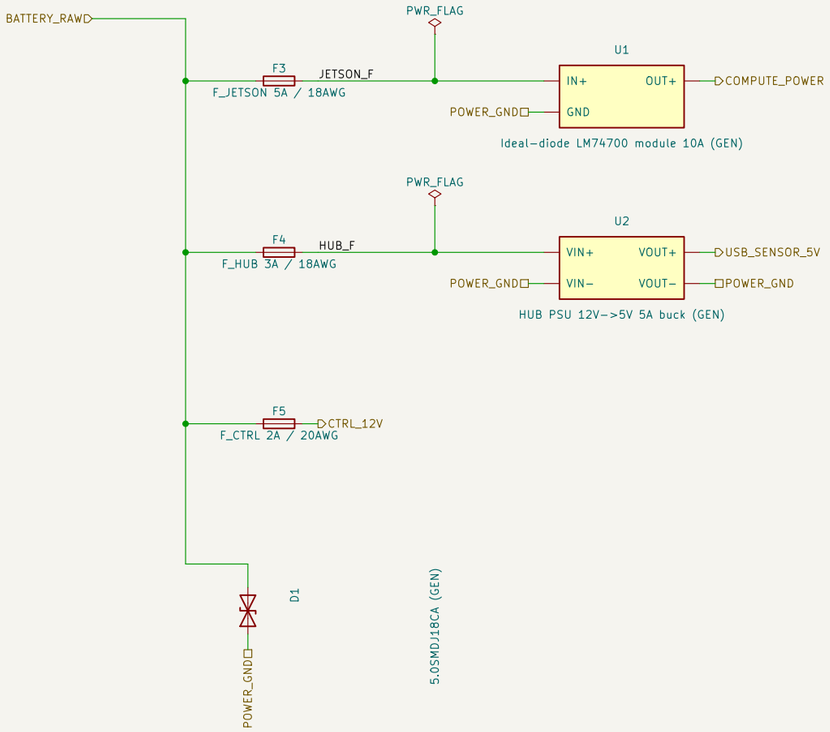
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.7 | Id: 2/10

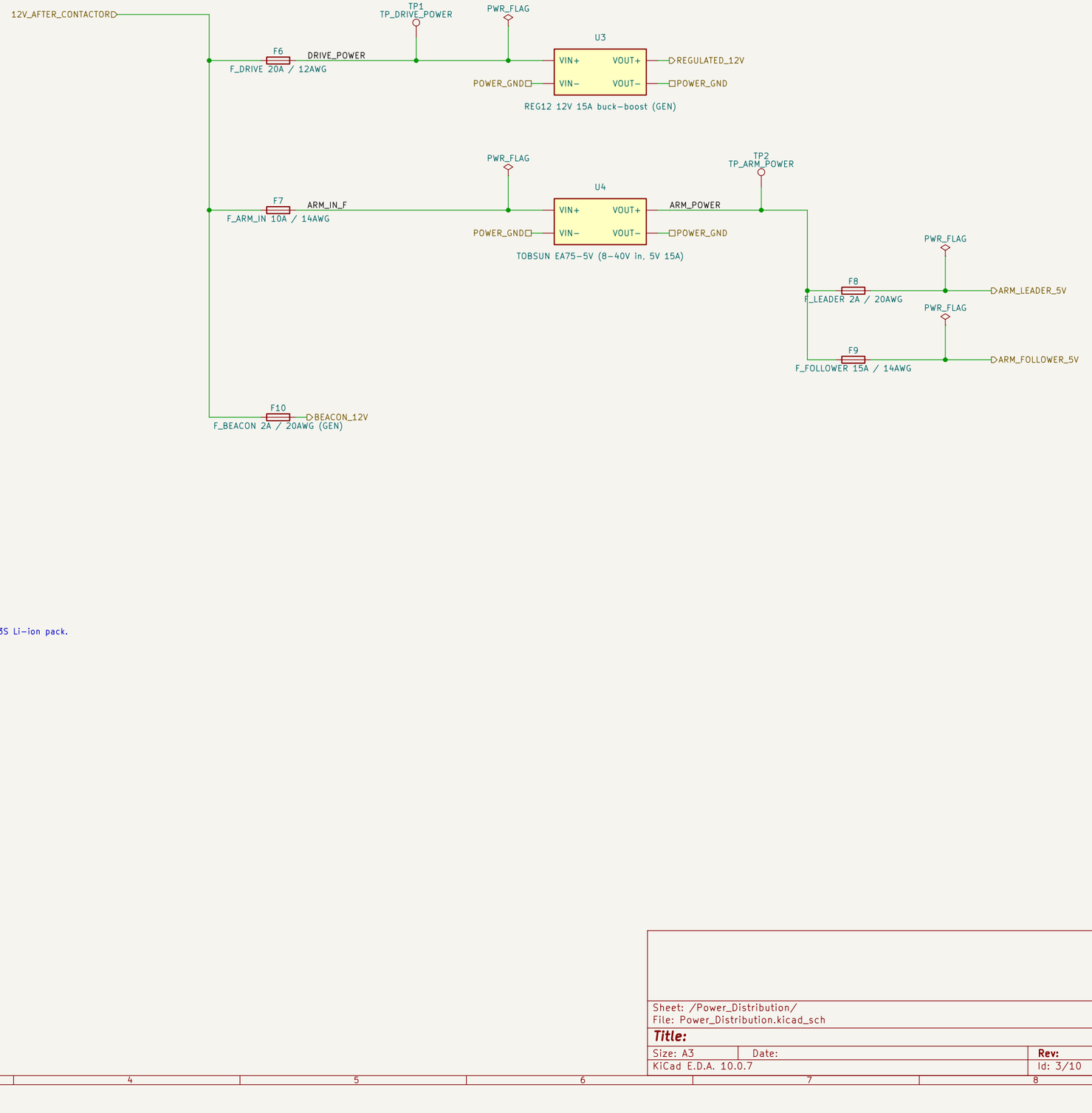
KIRO Wiring V2 | Power_Distribution

FB1 = unswitched fuse box (compute side) | FB2 = switched fuse box after contactor K1 (actuator side)

FB1 UNSWITCHED (not cut by E-stop)



FB2 SWITCHED (cut by E-stop via K1)



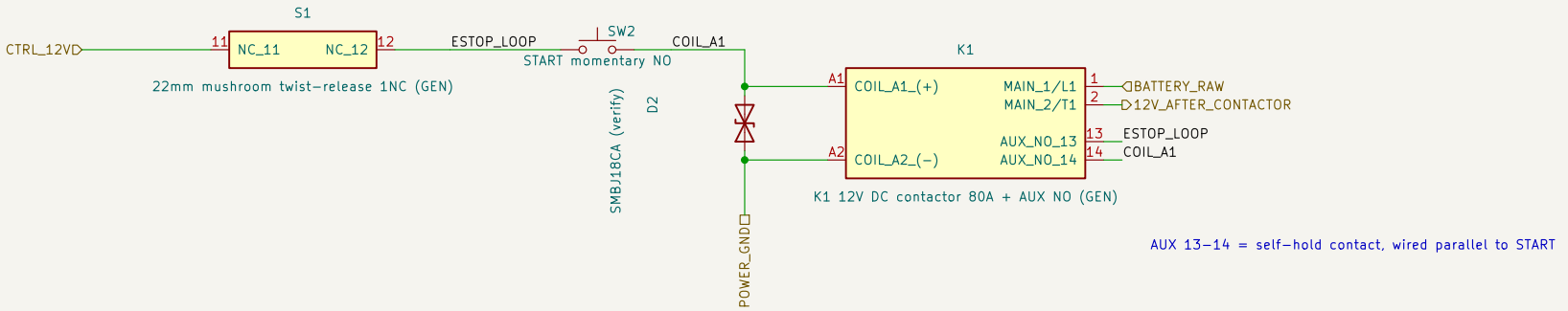
NOTES

1. Every branch fuse sits in FB1/FB2 at the start of its wire. Wire gauge shown in each fuse value; fuse <= wire and connector rating.
2. REG12 (REGULATED_12V) is required for lead-acid or LiFePO4: ST3215 limit 12.6 V, raw battery reaches 14.4-14.6 V. Omit only for a 3S Li-ion pack.
3. U1 reverse-polarity stage protects the Jetson branch. High-current branches rely on keyed J_BATT + F_MAIN.
4. TVS_BUS 18 V stand-off: above LiFePO4 14.6 V charge voltage (GEN - recheck if chemistry differs).
5. Pre-charge not fitted: generic REG12 input capacitance is small for an 80 A contactor. Re-check with the real REG12.
6. Arm 5 V rail: EA75-5V as in V1. Follower servo type on V2 (7.4 V or 12 V) is TBD - 5 V is safe for both.
7. HUB PSU stays on the unswitched side so cameras, LiDAR and adapter logic keep working after an E-stop.
8. #FLG = ERC power flags only (rail enters through a passive fuse). Not physical parts.

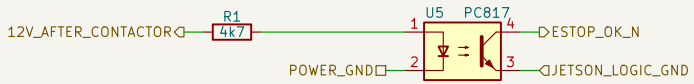
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File: Power_Distribution.kicad_sch		
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/10

KIRO Wiring V2 | Safety_EStop_and_Contactor

Hard-wired self-holding contactor loop. No software in the safety path.



AUX 13-14 = self-hold contact, wired parallel to START



ESTOP_OK_N to Jetson: LOW = 12V_AFTER_CONTACTOR present.

Opto LED on POWER_GND side; transistor on Jetson logic side.

Pull-up and 74LVC1G17 buffer: Jetson_and_GPIO sheet.

Broken wire -> pull-up -> HIGH -> software treats as E-stop.

E-STOP BEHAVIOUR

Press mushroom -> NC 11-12 opens -> coil current stops -> K1 opens -> 12V_AFTER_CONTACTOR = 0 V (wheels, arms, light).

Broken loop wire or blown F_CTRL -> same drop-out. Releasing the mushroom does not re-close K1 (AUX 13-14 open).

Restart: twist-release mushroom, then press START deliberately. COMPUTE_POWER is upstream of K1: Jetson stays up.

D2 bidirectional TVS clamps coil kick-back without the slow release of a plain flyback diode. Verify vs coil datasheet.

GENERIC: E-stop 1NC block 11-12; K1 12 V coil <= 1.5 A, main 80 A DC, AUX NO 13-14. Replace with real part data.

If K1 has no AUX NO: add a 12 V relay, coil parallel to K1 coil, and use its NO contact for the self-hold.

Sheet: /Safety_EStop_and_Contactor/
File: Safety_EStop_and_Contactor.kicad_sch

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KiCad E.D.A. 10.0.7 | Id: 4/10

KIRO Wiring V2 | Jetson_and_GPIO

NVIDIA Jetson Orin Nano Super Developer Kit, reference carrier: J16 DC jack, J12 40-pin header, J6/J7 USB

J12 HEADER FACTS (NVIDIA carrier spec)

All J12 signals 3.3 V. Non-I2C pins go through TI TXB0108 level translators: drive only +/-20 uA.

External pull-ups/pull-downs on these pins must be > 50 kOhm. Never load a J12 pin directly.

Therefore every GPIO here meets a 74LVC1G17 buffer (CMOS input, push-pull output).

Pins 29/31/7 are PROPOSED. Pinmux, boot state and state after cleanup: bench-verify (TBD).

J16: 9-20 V, 5.5 x 2.5 mm, centre positive, max 3.5 A. Raw battery 10-14.6 V is inside range.

Do NOT use hardware PWM for BEACON_HOLDOFF_PULSE: it must stop when software freezes.

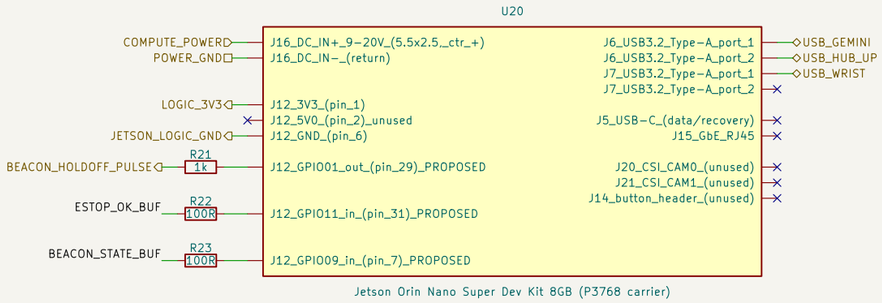
USB PORT ASSIGNMENT

J6 port 1: Orbbec Gemini 335 (USB 3.2 Gen1, bundled cable, never via hub)

J6 port 2: powered USB hub upstream (LiDAR + 3x CH343 servo adapters)

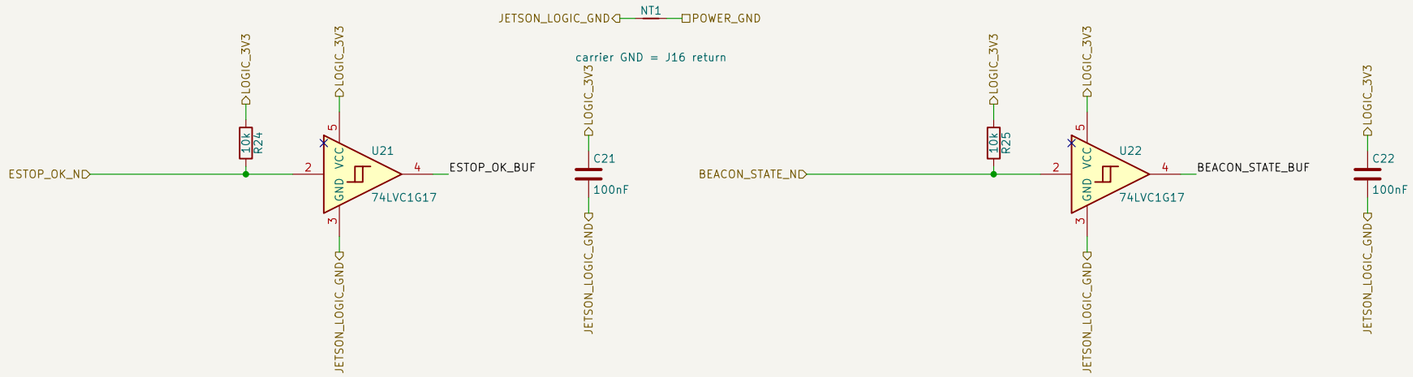
J7 port 1: Suyin wrist webcam. J7 port 2: spare. Each J6/J7 stack: 3 A VBUS total.

J15 GbE, J5 USB-C, J20/J21 CSI, J14 buttons: present on the carrier, not used in V2 (no-connect).



R21 1k: GPIO -> buffer input (light sheet). R22/R23 100R: buffer output -> TXB0108 B-side (overdrive).

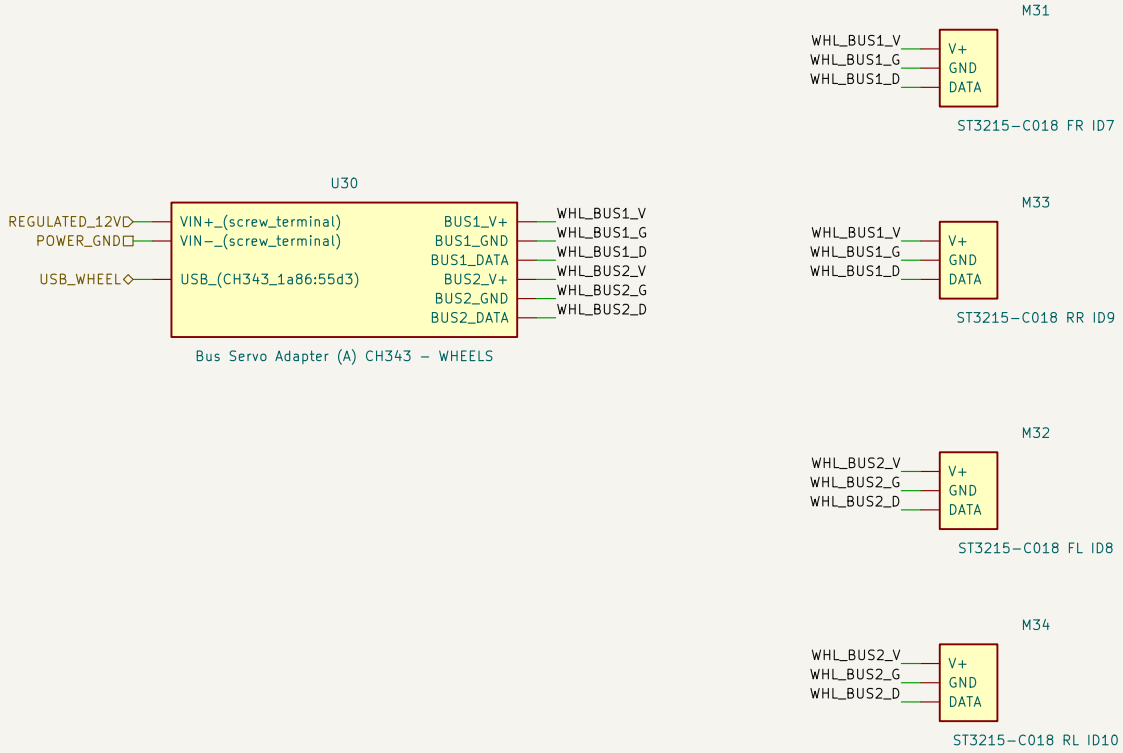
NT1: J12 GND and J16 DC return are the same carrier-board ground. No extra wire to POWER_GND.



Sheet: /Jetson_and_GPIO/	
File: Jetson_and_GPIO.kicad_sch	
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KIRO Wiring V2 | Wheel_Drive

4 x Feetech/Waveshare ST3215-C018 smart serial servos on one CH343 bus adapter (no generic motor driver)



Chain 1 (BUS1): adapter -> FR ID7 -> RR ID9. Chain 2 (BUS2): adapter -> FL ID8 -> RL ID10. Max two servos per Molex 5264 contact (~3 A).

Supply: REGULATED_12V (ST3215 6-12.6 V; V1 Max_Voltage_Limit 14.0 V). Adapter powered via screw terminal, not its DC jack.

Current: idle ~1.2 A total, design 6 A, all four stalled 10.8 A (2.7 A each). Feed 14 AWG to screw terminal; 20 AWG servo leads.

Servo cable: 3-pin Molex 5264 (V+, GND, DATA). Physical pin order: verify against Waveshare ST3215 wiki before crimping (TBD).

Data: half-duplex TTL 1 Mbps. USB_WHEEL via powered hub. Servo GND returns to the adapter, never through the USB cable shield.

E-stop: K1 removes REGULATED_12V -> servos unpowered, no holding torque, robot coasts. Speed feedback over bus = ROBOT_MOVING.

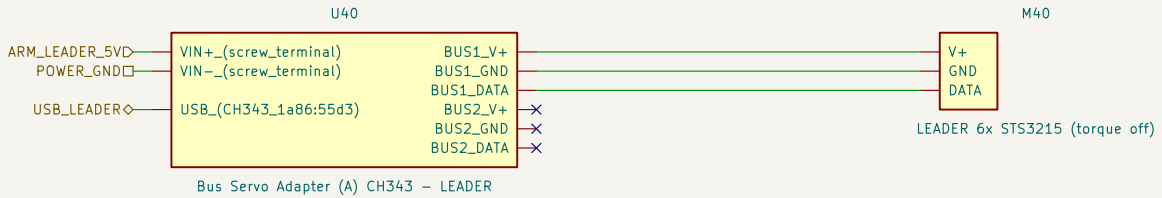
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File: Wheel_Drive.kicad_sch

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KiCad E.D.A. 10.0.7 | Id: 6/10

KIRO Wiring V2 | Leader_Follower_Arms

LeRobot leader + follower arms, each on its own CH343 bus adapter and its own fused 5 V branch



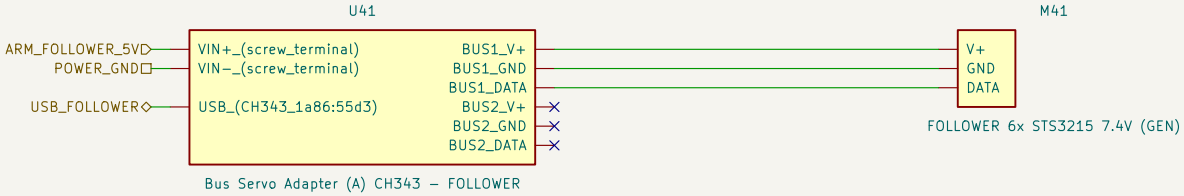
LEADER: F_LEADER 2 A, 20 AWG. Torque off (hand-guided). ~0.2 A typ, < 1 A peak (V1-doc). After K1 as requested.

FOLLOWER: F_FOLLOWER 15 A, 14 AWG. ~1–2 A typ, ~6 A design, ~9 A peak at 5 V (V1-doc, verify on V2).

Voltage: V1 follower = 7.4 V STS3215 (trip 8.0 V) -> keep 5 V. A 12 V STS3215 variant would need a new rail (Q2, TBD).

E-stop: K1 cuts EA75-5V input -> both arms limp. Follower falls under gravity: keep people clear of the arm envelope.

Bus 2 of each adapter unused (no-connect). USB to powered hub. Power return by its own wire to the star point.



Adapter feed via screw terminal. Vendor nominal input 9–12.6 V; V1 runs these adapters from 5 V successfully (V1-measured) — keep, verify.

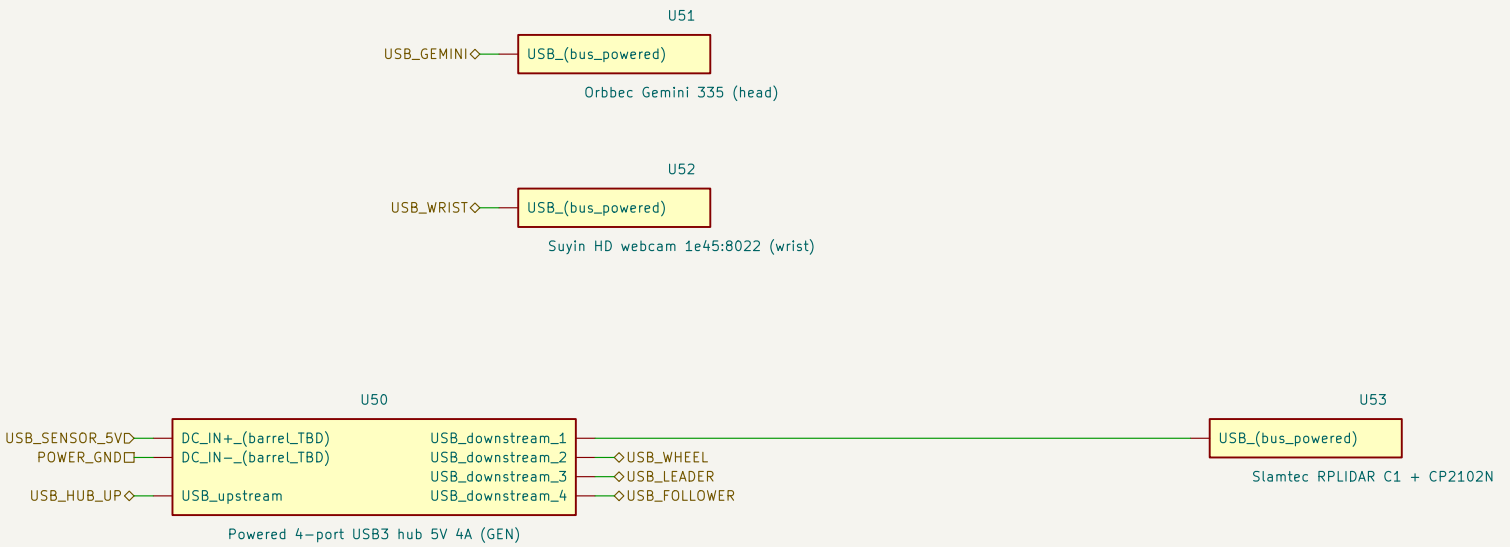
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File: Leader_Follower_Arms.kicad_sch

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Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 7/10

KIRO Wiring V2 | USB_Hubs_Cameras_and_LiDAR

Only the installed hub, two cameras and one LiDAR. No other sensors.



DEVICE TABLE

Gemini 335 | USB bus 5 V | 896 mA max, <6.5 W | USB 3.2 Gen1 | bundled cable | Jetson J6 port 1 direct

Wrist webcam | USB bus 5 V | needs 500 mA | USB 2.0 | camera cable | Jetson J7 port 1 direct

RPLIDAR C1 | 5 V via USB | 230 mA typ, peak TBD | UART 460800 inside, USB outside (CP2102N) | hub DS1

USB hub | DC in 5 V 4 A, barrel 5.5 x 2.1 mm, centre positive (GEN − check against real hub label)

Hub barrel: if the real hub is 3.5x1.35 or 5.5x2.5 mm, change the plug; never trust "5 mm" alone.

Shielding: use shielded USB cables; route away from servo and 12 V harnesses. No motor current returns via USB.

LiDAR start−up (V1): plug sensor cable into adapter with USB unplugged, then connect USB. Motor spin−up after host opens port.

Sheet: /USB_Hubs_Cameras_and_LiDAR/
File: USB_Hubs_Cameras_and_LiDAR.kicad_sch

Title:

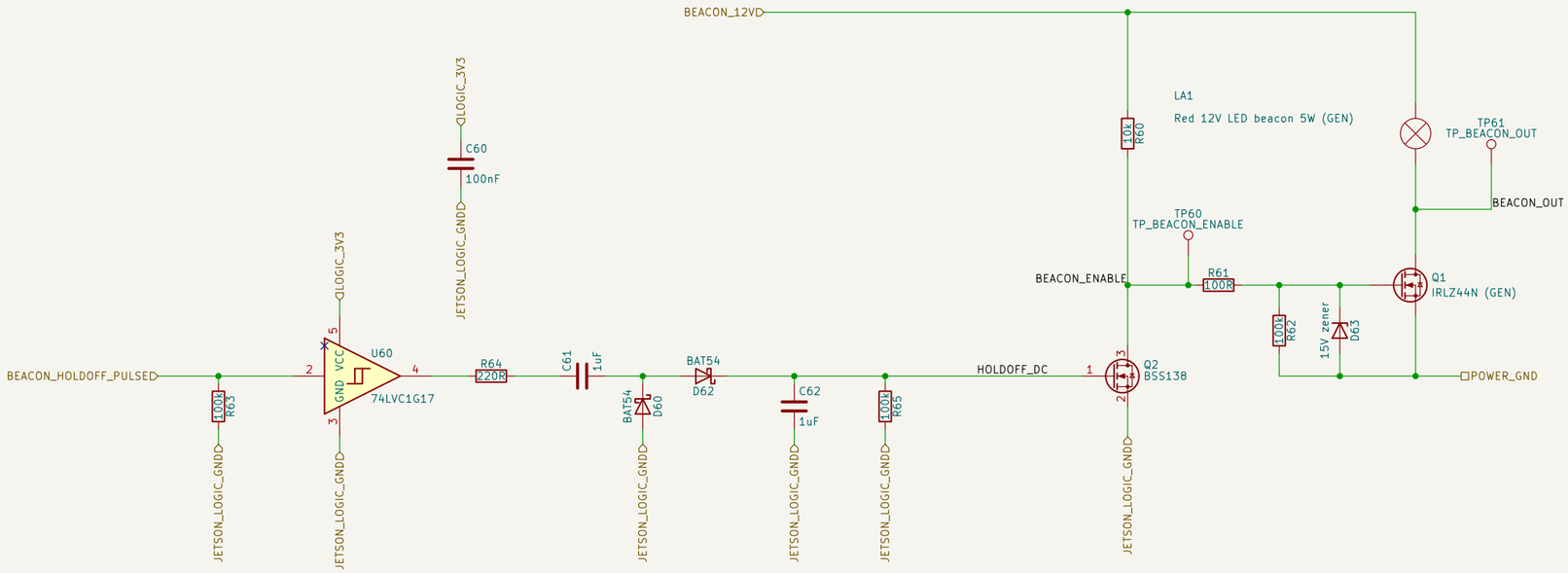
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 8/10

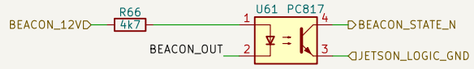
KIRO Wiring V2 | Red_Motion_Warning_Light

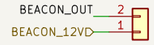
Fail-to-warning: light is ON whenever K1 is closed, unless the Jetson actively holds it OFF with a live pulse train

SOFTWARE MEANING: MOTION_WARNING_REQUEST active = stop toggling BEACON_HOLDOFF_PULSE = light ON

Light OFF only while the heartbeat thread toggles GPIO01 (J12 pin 29) at >= 100 Hz square wave. Frozen, crashed, booting or powered-off Jetson = light ON (if armed).







CHARGE-PUMP WATCHDOG (estimate, verify on bench)

Pulses through C61 and D60/D62 charge C62 to ~2.2-2.5 V -> Q2 on -> BEACON_ENABLE pulled low -> Q1 off -> light off.

Pulses stop (either level) -> C62 bleeds through R65: tau = 100k x 1uF = 0.1 s -> Q2 off within ~0.1-0.15 s -> R60 pulls gate up -> light on.

Gate drive - BEACON_12V x 100k/110k = 9-13 V. D63 clamps Vgs. Q1: Vds >= 30 V, Id >= 3x lamp current, Rds(on) specified at Vgs = 10 V.

R63 100k (> 50k rule): pin floats at boot -> buffer input low -> no pulses -> light ON while armed (fail-safe).

Feedback U61: LED across the lamp. Lamp energised -> BEACON_STATE_N LOW. Confirms drive, not filament - check lamp at bring-up.

POWER PATH: 12V_AFTER_CONTACTOR -> F_BEACON (Power_Distribution) -> BEACON_12V -> J60.1 -> lamp + ... lamp - -> J60.2 -> Q1 drain -> Q1 source -> POWER_GND

GENERIC: lamp 12 V 5 W (~0.42 A). F_BEACON 2 A (LED inrush), 20 AWG, J60 5 A, Q1 IRLZ44N (55 V, Rds 22 mOhm @10 V). Re-check when the lamp is measured.

Polarity: generic LED beacon is polar - red wire to J60.1 (+). TVS not required for a lamp/LED load.

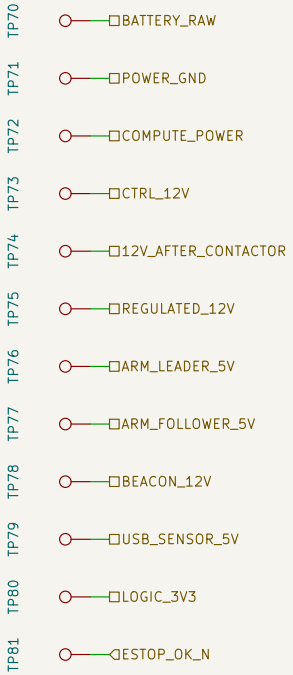
SEQUENCE 1 non-zero wheel cmd -> 2 stop pulses (request) -> 3 light on, confirm BEACON_STATE_N low (timeout 0.5 s -> refuse motion, log)

4 wait 0.5 s -> 5 forward wheel cmd -> 6 keep light on while any cmd != 0 -> 7 after zero cmd keep on 1.0 s -> 8 resume pulses (light off)

Sheet: /Red_Motion_Warning_Light/		
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KiCad E.D.A. 10.0.7		Id: 9/10

KIRO Wiring V2 | Connectors_and_Test_Points

Test-point panel on every cross-sheet rail, plus the connector schedule



CONNECTOR SCHEDULE (rating must exceed fuse; pinout verified before crimping)

J_BATT (J1) Anderson SB50, keyed, 50 A ..... GEN

J_CHG (J2) XT60 charger inlet ..... GEN

J16 Jetson 5.5 x 2.5 mm barrel, centre +, 3.5 A max . NVIDIA spec

Adapter VIN screw terminal (not the 5.5 x 2.1 jack) .. Waveshare

Servo bus Molex 5264 3-pin, use factory servo leads

Hub DC in 5.5 x 2.1 mm, centre +, 5 V 4 A ..... GEN

J_BEACON (J60) 2-pin 5 A, pin 1 = lamp + ... GEN

ESTOP / START panel wiring, 20 AWG, ferrules ..... GEN

K1 main M6 ring lugs, 80 A DC contact ..... GEN

Signal harness J12 pins 1/6/29/31/7 to buffer board, 2.54 mm Dupont, strain-relieved

TEST-POINT USE

Measure all rails against TP_POWER_GND at the star point. Local TPs:

TP1 DRIVE_POWER, TP2 ARM_POWER (Power_Distribution); TP60 BEACON_ENABLE,

TP61 BEACON_OUT (Red_Motion_Warning_Light). Strain relief on every harness exit.

Sheet: /Connectors_and_Test_Points/ File: Connectors_and_Test_Points.kicad_sch		
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 10/10